Jingwei Ni

Ph.D. Candidate · Distributional Alignment and Uncertainty for Trustworthy AI

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Research Vision

People instruct agents and write rubrics in natural language, which leaves a **fuzzy** and unverifiable part whose ground truth is a **distribution** of human judgments rather than a point answer. I believe that getting this part right, so that a model is distributionally aligned with people and knows when it is uncertain, is key to teaching AI taste, preventing reward hacking, and aligning AI beyond the verifiable. My PhD work covers two foundations: **judgment with fuzzy criteria**, evaluating and improving how LLMs handle human disagreement and building human-AI collaborative annotation tools, and **uncertainty quantification** that exposes a model's confidence about its epistemic states.

Research Overview

Judgment with fuzzy criteria: evaluating LLMs against human disagreement
LLM judges inherit the fuzzy part of a rubric, so I test models against **distributions** of human judgments rather than a majority label. My EACL 2026 work shows that LLMs collapse legitimate annotator disagreement into one overconfident answer, and that RLVR-style reasoning training makes this worse. DIRAS learns **calibrated** judgments under nuanced natural-language criteria and distills them into small models.

Judgment with fuzzy criteria: human-AI collaborative annotation tools
Judgment under fuzzy criteria needs people in the loop, and the tool decides where their judgment goes. Co-DETECT lets an expert refine a rough codebook: it uses **annotation uncertainty** to surface edge cases the codebook does not cover, induces generalizable rules for them, and leaves the expert in charge of what enters the codebook. AFaCTA calibrates LLM annotators on a small human set and routes items between people and models, so validated labels scale into benchmarks.

Uncertainty quantification: exposing a model's epistemic states
Honest clarification requires that a model knows where it is unsure. ReProbe reads a frozen LLM's **internal states** to estimate the credibility of each reasoning step, enabling efficient verification and test-time search without a large process reward model. BELT reads a model's **beliefs** about its current and future answer from the same hidden states and fits a small stopping policy on them, so one probe serves different accuracy-cost trade-offs and transfers from math to coding and agentic tasks.

Industry & Collaborative Research Experience

Qwen Business Unit of Alibaba, Research Intern — Jan 2026 – Jun 2026

Agent self-evolution, test-time learning, and RLVR/RLHF post-training.

- Developed **Trace2Skill** (**EMNLP 2026 Main**), a verifier-grounded framework that distills successful and failed agent trajectories into reusable skills for improving future test-time behavior.
- Investigated **skill-based agent self-evolution** across inference-time experience accumulation and combined **RLVR/RLHF post-training**.
- Mentored **Skill-RM**, which uses agent skills to unify heterogeneous evaluation criteria; and **GD²PO**, which addresses conflicts among multiple rewards during policy optimization.

Swiss AI Initiative, Research Contributor, Apertus Open LLMs — Jun 2025 – Aug 2025

- Contributed trustworthiness and hallucination-reduction post-training to the Apertus series, a fully open and compliant suite of LLMs. Apertus report.

Evaluating Evaluations Coalition, Research Contributor, When AI Benchmarks Plateau — Jul 2025 – Jan 2026

- Contributed to an ICML 2026 study of saturation across 60 language-model benchmarks and design choices for more durable evaluation. Paper.

Education

PhD **ETH Zürich**, Computer Science — Zürich, Switzerland — 2023 – present

- Advisors: Profs. Elliott Ash, Mrinmaya Sachan, and Markus Leippold
- Research: distributional alignment to fuzzy human criteria and uncertainty quantification
- Expected completion: June 2028

MSc **University College London (UCL)**, Data Science and Machine Learning — London, UK — 2021 – 2022

- Distinction

BEng **University of Hong Kong**, Computer Science — Hong Kong, 2017 – 2021
- First Class Honours

Selected Publications

ReProbe: Efficient Test-Time Scaling of Multi-Step Reasoning by Probing Internal States of Large Language Models — ACL 2026 (Oral)
Jingwei Ni, et al.
aclanthology.org/2026.acl-long.536

Same Beliefs, Different Policies: Learning Reusable Representations for Early Stopping in LLM Reasoning — arXiv 2026 (Preprint)
Jingwei Ni, et al.
arxiv.org/abs/2610.30450

Trace2Skill: Distill Trajectory-Local Lessons into Transferable Agent Skills — EMNLP 2026 (Main)
Jingwei Ni, et al.
arxiv.org/abs/2603.25158

Can Reasoning Help Large Language Models Capture Human Annotator Disagreement? — EACL 2026 (Oral)
co-first author
Jingwei Ni, et al.
aclanthology.org/2026.eacl-long.3

DIRAS: Efficient LLM Annotation of Document Relevance for Retrieval-Augmented Generation — NAACL 2025 (Long)
Jingwei Ni, et al.
aclanthology.org/2025.naacl-long.271

Co-DETECT: Collaborative Discovery of Edge Cases in Text Classification — EMNLP 2025 (Demo)
Chenfei Xiong*, *Jingwei Ni**, et al.
* *co-first author*
aclanthology.org/2025.emnlp-demos.25

Towards Faithful and Robust LLM Specialists for Evidence-Based Question-Answering — ACL 2024 (Long)
* *co-first author*
Tobias Schimanski*, *Jingwei Ni**, et al.
aclanthology.org/2024.acl-long.105

AFaCTA: Assisting the Annotation of Factual Claim Detection with Reliable LLM Annotators — ACL 2024 (Long)
Jingwei Ni, et al.
aclanthology.org/2024.acl-long.104

Invited Talks

Hard-to-Verify Reasoning, AI4Law Workshop @ ICML 2026 — 2026
Invited talk and panel contribution on hard-to-verify reasoning for legal AI. Workshop.

Teaching Experience

Head Teaching Assistant, *Language Models for Law and Social Science* — ETH Zürich, 2023 – 2025
Graduate course by Prof. Elliott Ash. Head TA for the 2023, 2024, and 2025 editions.

Mentoring & Supervision

Skill-RM, Qwen
Research mentorship on using agent skills to unify heterogeneous evaluation criteria. Mentee: Tao Chen. Preprint.

GD²PO, Qwen
Research mentorship on resolving conflicts among multiple rewards during policy optimization. Mentee: Haotian Liu. **EMNLP 2026 Findings**. Paper.

Understanding Failures in LLM Reasoning by Learning Structured Representations of Chain-of-Thought
Research mentorship. Mentee: Tommaso Felice Banfi. **EMNLP 2026 Findings**. Paper.

Minjing Shi (MSc, ETH Zürich)
Tackling the Root of Misinformation by Teaching Laypeople about Logical Fallacies via Socratic Questioning and Critical Argumentation, published at **ACL 2026**.

Adam Rahmoun (MSc, ETH Zürich)
Unlocking LLM Legal Reasoning with IRAC-Constrained Chain-of-Thought, published at **AI4Law @ ICML 2026**.

Qing Dai (MSc, ETH Zürich)
From Dataset to Optimization: A Benchmarking Framework for Information Retrieval in the Particle Accelerator Domain, with code and data released in 2025.

Academic Service

Core Organizer, ClimateNLP Workshop @ ACL 2024 – 2026
Organized the ACL 2024 and ACL 2025 editions and the 2026 continuation, connecting academia and industry around NLP for climate-change science. Workshop site.

Reviewer
ACL Rolling Review (ARR), 2023–2026.

Honors & Awards

- Oral presentations at ACL 2026 and EACL 2026
- MSc awarded with Distinction, UCL (2022)
- First Class Honours, University of Hong Kong (2021)

Technical Skills

ML & NLP: Uncertainty quantification, calibration to human disagreement, human-AI collaborative annotation, LLM reasoning and test-time scaling, agent self-improvement, SFT, RLHF, RLVR, retrieval-augmented generation, and large-scale evaluation.

Engineering: Python, PyTorch, distributed training, large-scale HPC and Slurm experiment pipelines, and data pipelines.

References

Prof. Elliott Ash
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Prof. Mrinmaya Sachan
Department of Computer Science, ETH Zürich (PhD advisor) · mrinmaya.sachan@inf.ethz.ch

Prof. Markus Leippold
Department of Finance, University of Zürich (PhD advisor) · markus.leippold@df.uzh.ch